Heydar Soudani

ML Engineer · LLM Training & Post-training · RAG & Agents · Uncertainty Quantification

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Based in the Netherlands · Available immediately

Summary

Final-year PhD candidate in Computer Science at Radboud University training, building and evaluating LLM systems for **retrieval-augmented generation (RAG)**, **deep research agents** and **uncertainty quantification**. At **Thomson Reuters Labs**, built agentic search for legal research and a controller that cut agents' search calls by 14 on average with up to 10% higher accuracy (EMNLP 2026); at **KPN**, brought uncertainty estimation to a customer-facing RAG chatbot. Hands-on with LLM fine-tuning (full, LoRA / QLoRA), RL post-training, synthetic data generation, multi-GPU jobs on AWS and HPC, and benchmark construction. Published at EMNLP, SIGIR, ACL, ICTIR, SIGIR-AP and ACM Computing Surveys.

Experience

Thomson Reuters Labs *Nov 2025 – May 2026*

Data Scientist Intern (Applied AI Research) Zug, Switzerland (Remote)

- **Agentic search for legal research**: replaced a single-step retrieval pipeline with a modular **agentic search system** (planner, query decomposer, retriever, reranker, summarizer) to raise recall. Iterated on prompts and component wiring with the **product team**, evaluating each change in the product.
- **Stagnant reasoning** (*EMNLP 2026*): showed across **7 deep research agents** and 3 datasets that recall plateaus after a few search calls while agents keep searching. Built **RAAC**, a controller that uses 4 unsupervised trajectory signals to continue, intervene or stop: **14 fewer search calls** on average and up to **10% higher accuracy** on BrowseComp-Plus, and **58% fewer iterations** with higher recall on LegalSearch.
- **Query decomposition & reformulation study**: benchmarked decomposition and reformulation methods (HyDE, Query2doc, Rewrite-Retrieve-Read, ThinkQE) with **8 rerankers** (MonoT5, RankLLaMA, Rank1, Rank-R1, Qwen3, RankGPT, RankZephyr) and **rank fusion** (RRF, interleaving, bandits). Found that one-shot methods are capped by the LLM's own knowledge, which motivated agentic search.
- **Infrastructure**: **AWS** (SageMaker multi-GPU jobs, Bedrock), **Langfuse** trajectory tracing, Claude and GPT APIs, and open-weight Qwen models served with **vLLM**.

KPN *May 2025 – Oct 2025*

Data Scientist Intern Amsterdam, the Netherlands

- **Uncertainty for a production chatbot**: applied my axiomatic uncertainty method (*ACL Findings 2025*) to KPN's customer-facing **RAG chatbot**, which answers from a **knowledge graph** and text, to score its confidence in each answer and flag unreliable responses.

Radboud University *Feb 2023 – Present*

PhD Candidate, Trustworthy Generative Information Access Nijmegen, the Netherlands

- **Uncertainty for search agents** (*SIGIR 2026*): built **R2C**, which perturbs an agent's reasoning-and-search trajectory and scores how consistent its answers stay. Across 5 agents: **+5% AUROC**, about 5% better abstention F1, and about 7% higher exact match when used to pick between models.
- **Uncertainty in RAG** (*ACL Findings 2025*): defined **5 axioms** for how uncertainty should change under retrieval, showed existing methods meet only 2, and derived a **calibration function** that brought RAG AUROC to or above the no-retrieval level.
- **Multimodal uncertainty** (*ICTIR 2026*): co-developed **LeMUQ**, a **fine-tuned RoBERTa** scorer that reads a VLM's token probabilities with the image or retrieved context removed, improving AUROC by **0.038** on average.
- **Fine-tuning vs. RAG** (*SIGIR-AP 2024*): compared **full fine-tuning**, **QLoRA**, synthetic QA generation and RAG on **12 LMs** (80M–11B). RAG won on long-tail entities, and the proposed **Stimulus RAG** beat fine-tuning with no training cost.
- **Total Recall QA** (*SIGIR 2026*): co-built a **verifiable benchmark** for deep research agents, with about 2.3K queries over a 57.7M-document corpus. Even GPT-5.2 with the gold documents reaches only 57% exact match.

- **Synthetic conversational data** (*ACM Computing Surveys 2026*): led a survey of **synthetic multi-turn dialogue generation** for open-domain, task-oriented and information-seeking systems. Proposed a general framework covering seed data creation, utterance generation and **quality filtering**.
- **Tutorials**: Data Augmentation for Conversational AI, 2 editions (*CIKM 2023*, *WebConf 2024*).

Atrovan *Dec 2017 – Jun 2022*
Web Developer (part-time) Tehran, Iran

- **Products**: built and maintained **4 production web products** for industrial and individual users: smart metering, fleet management, home automation and building management.
- **Engineering**: **React.js**, JavaScript, **REST APIs**, **PostgreSQL**, **Docker**, **Kubernetes** and Git, in a 10+ person **Scrum** team, contributing to **system design**.

Polytechnic of Tehran *Jan 2016 – Oct 2017*
Research Assistant, Control of Multi Vehicle Systems Lab Tehran, Iran

- **Embedded systems and vision**: **vision-based localization**, AVR / Raspberry Pi programming and PCB design.

Selected Publications

[1] **EMNLP 2026**. When Deep Research Agents Stagnate: Enhancing Reasoning with Retrieval-Aware Agent Control
[2] **SIGIR 2026**. Uncertainty Quantification for Retrieval-Augmented Reasoning
[3] **SIGIR 2026**. Total Recall QA: A Verifiable Evaluation Suite for Deep Research Agents
[4] **ICTIR 2026**. Uncertainty Quantification for Multimodal Retrieval Augmented Generation
[5] **ACM Computing Surveys 2026**. A Survey on Recent Advances in Conversational Data Generation
[6] **ACL Findings 2025**. Why Uncertainty Estimation Methods Fall Short in RAG: An Axiomatic Analysis
[7] **SIGIR-AP 2024**. Fine Tuning vs. Retrieval Augmented Generation for Less Popular Knowledge
[8] **COLING 2022**. Persian Natural Language Inference: A Meta-learning Approach

Technical Skills

Languages: Python, SQL, Bash, JavaScript, C/C++
ML / DL: PyTorch, Hugging Face (Transformers, PEFT, Datasets), scikit-learn, TensorFlow, NumPy, pandas
LLM training & post-training: SFT, full fine-tuning, LoRA / QLoRA, RL post-training (GRPO, PPO, DPO, RLVR), reward design, synthetic data generation
LLM inference & models: vLLM, LiteLLM, OpenRouter, AWS Bedrock; Claude, GPT, Qwen, GLM, GPT-oss, vision-language models
Agents & RAG: deep research agents (ReAct, Search-R1, Tongyi-DR), agentic search, tool use, RAG over text and knowledge graphs, query decomposition & reformulation, dense & sparse retrieval (BM25, SPLADE, E5, Qwen3-Embedding), LLM rerankers, rank fusion, FAISS, LangChain / LangGraph, prompt engineering
Evaluation: uncertainty quantification & calibration (AUROC), LLM-as-judge, benchmark construction, retrieval evaluation, ablation studies, trajectory & failure analysis
MLOps & cloud: AWS (SageMaker, Bedrock), multi-GPU training & inference, HPC / Slurm, Langfuse, Weights & Biases, Docker, Kubernetes
Software engineering: Git, Linux, REST APIs, PostgreSQL, React.js, system design, Scrum
AI-assisted development: Claude Code (daily; custom skills, hooks and MCP servers), GitHub Copilot, Cursor

Education

PhD Computer Science — Radboud University, 2023–2027 (expected)
MSc Computer Engineering — Sharif University of Technology, 2022
BSc Electrical Engineering — Polytechnic of Tehran, 2019
BSc Computer Engineering (minor) — Polytechnic of Tehran, 2018

References

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